PRINCIPLES OF AI · FALL 2026 · FLCS #2

Real-Time Facial Expression Recognition

ResNet50-based visible facial-expression classification



Team Group #3
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Team contributions

Presentation responsibilities and assigned sections.

MEMBER	PRIMARY TASK	DELIVERABLE	SECTION
Ahsan Ullah ID: 202680300	Introduction, project objective and problem context	Slides 1–2	01–02
Mohammed Sahed Saddam ID: 202680290	W5H1 analysis: what, why and who	Slides 3–4	03–04
Thushyanth Jegatheesan ID: 202680254	CNN fundamentals and ResNet50 architecture	Slides 5–6	05–06
Izhar Ullah ID: 202680274	FER2013 dataset, preprocessing and training	Slides 7–8	07–08
MOJ SHENGJUN ID: 202680067	Model evaluation, metrics and confusion matrix	Slide 9	09
Gulaiym Ibraimova ID: 202680263	Live webcam demonstration and conclusion	Slide 10	10

TEAM 6 MEMBERS

Facial Expressions as Visual Data

Human faces contain visual patterns that computer vision can analyze. Our project uses deep learning to classify seven visible facial-expression categories in real time.



01

Detect a face in live video

02

Classify 7 visible expressions

03

Respond in real time

04

Report confidence responsibly

angry · disgust · fear · happy · sad · surprise · neutral

W5H1: What, Why & Who

WHAT

A camera-to-label system that detects a face and predicts one of seven visible expression classes.

WHY

Fast, repeatable visual classification can support HCI research, accessibility studies and learning tools.

WHO

Researchers, developers and educators — with informed consent, clear boundaries and human oversight.

Visible expression ≠ internal emotional state

W5H1: Where, When & How

A local, low-latency loop for classroom demonstration and prototyping.



WHERE local device / lab WHEN every frame HOW detect → transform → infer → stabilize → display

How CNNs read facial features

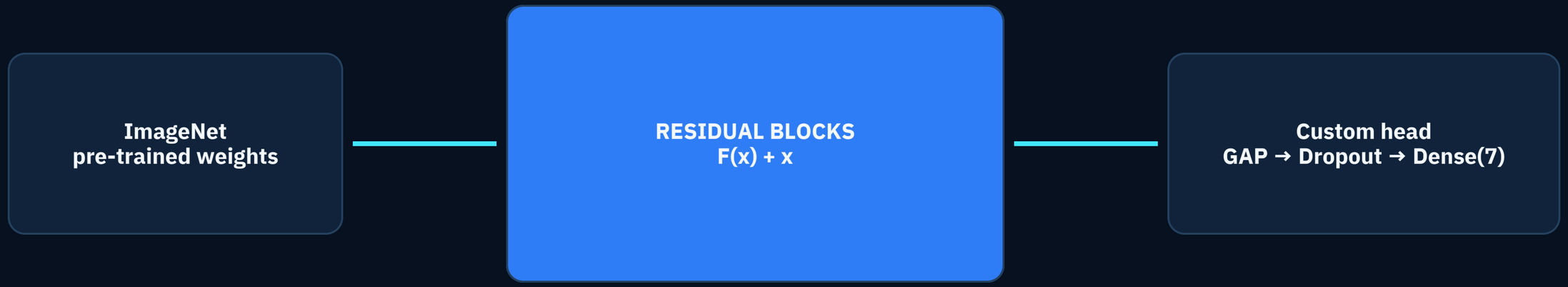
Each stage converts pixels into increasingly useful patterns.



Small filters learn local cues — eyebrow angle, eye openness, mouth shape. Deeper layers combine them into expression-level evidence.

Why ResNet50?

“50” refers to the depth of the original architecture: 49 convolutional layers plus its final classifier. We use the pretrained convolutional backbone and add our own 7-class head.



skip connections stabilize depth

reuse general visual features

fine-tune upper layers only

Dataset: FER2013

FER2013 facial images are resized to 224 × 224 RGB; the training folder is split into stratified training and validation sets.



35,887
total images

24,402
training (85%)

7,178
test

224 × 224
model input

STRATIFIED SPLIT 28,709 train-pool images → 24,402 training + 4,307 validation | AUGMENTATION flip · rotation · zoom · translation · contrast

Our model & training process

224×224×3 → Augmentation → ResNet50 (ImageNet, include_top=False) → GAP → BatchNorm → Dense(256, ReLU) → Dropout(0.40) → Dense(128, ReLU) → Dropout(0.25) → Dense(7, Softmax)

STAGE 1

Train classifier head

Frozen backbone · Adam 0.001 · up to 12 epochs

STAGE 2

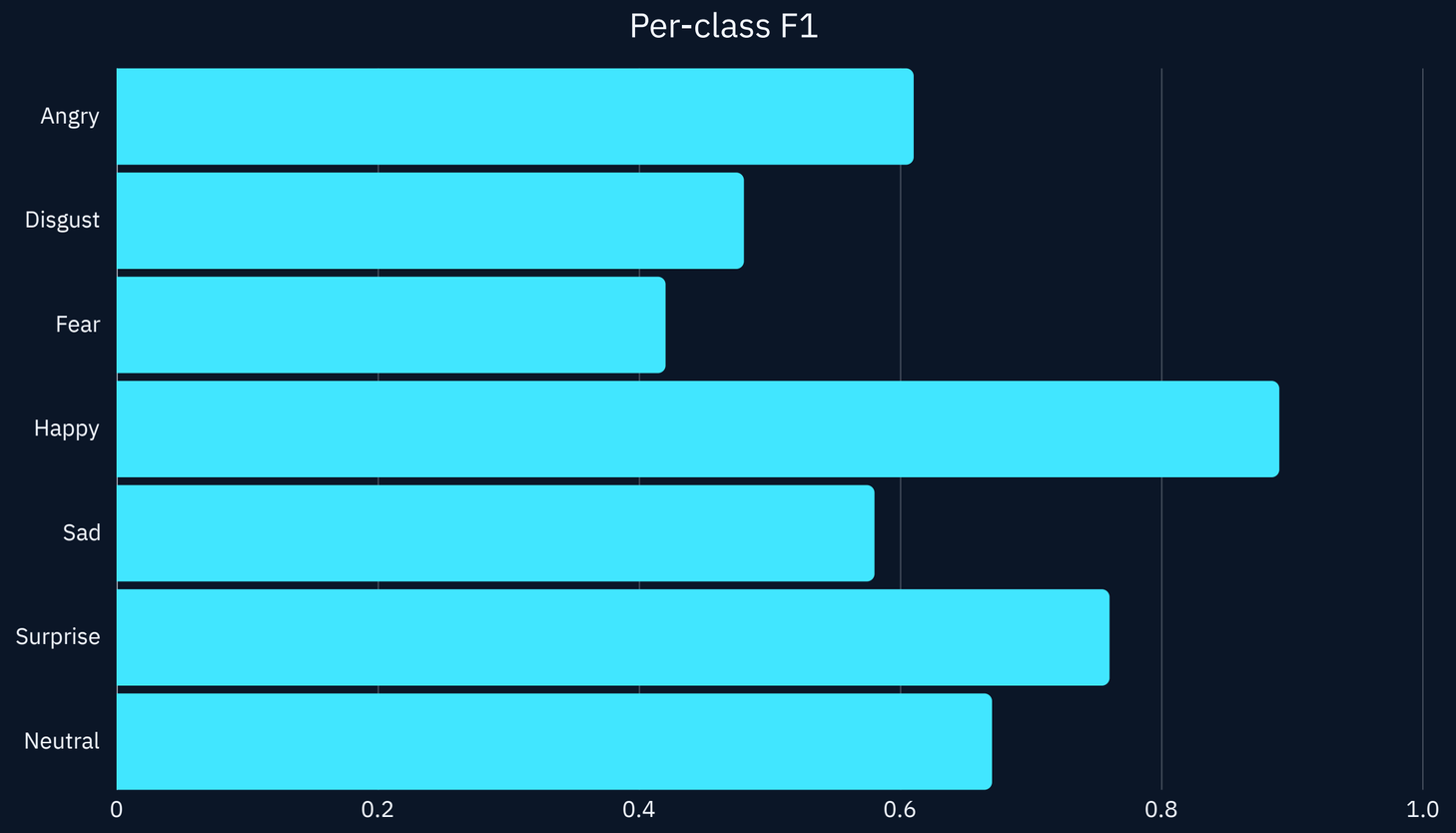
Fine-tune last 30 layers

BatchNorm frozen · Adam 0.00001 · up to 25 epochs

TensorFlow / Keras • OpenCV • NumPy • scikit-learn

Test results & evaluation

Independent test set: 7,178 images. The original ResNet50 model achieved 64.96% accuracy, 62.91% macro F1 and 64.54% weighted F1.



64.96%
test accuracy

62.91%
test macro F1

64.54%
test weighted F1

CLASS RESULTS happy F1: 84.16% • fear F1: 46.66%

Live webcam demo

Detect → classify → smooth → display — with confidence-aware fallback.



1 OpenCV face detection

2 5-frame probability smoothing

3 Show label only if confidence ≥ 0.40

4 Limitations: lighting, pose, occlusion, bias

5 Use only with consent and human oversight

References

Core sources for the model, dataset and implementation.

01 He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. CVPR.

02 Goodfellow, I. J., et al. (2013). Challenges in representation learning: A report on three machine learning contests.

03 TensorFlow / Keras. ResNet50 and transfer-learning documentation. tensorflow.org

04 OpenCV. Cascade classification and video capture documentation. docs.opencv.org

05 FER2013 dataset source: [INSERT OFFICIAL / COURSE-APPROVED URL]

Questions?